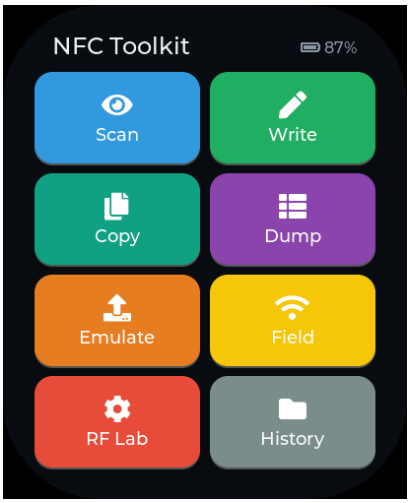
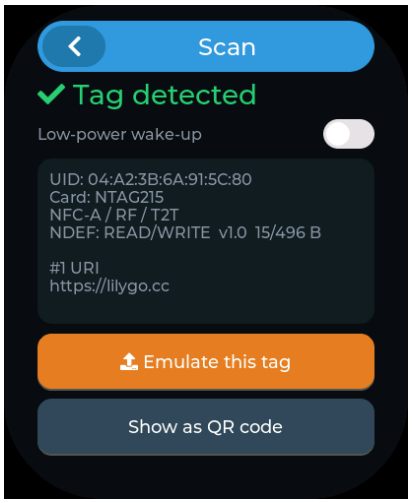
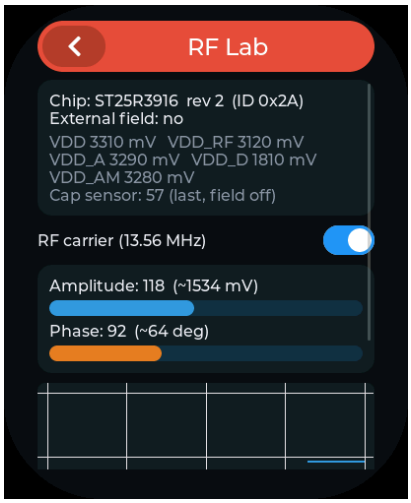
NFC Toolkit

for the LilyGO T-Watch Ultra (ST25R3916)

User guide: installing, using every tool, and what it can read, write and emulate







The T-Watch Ultra contains an ST25R3916, the same NFC front-end STMicroelectronics uses in its development kits. It can read and write tags, act as a tag itself, sense other NFC readers and measure its own antenna. NFC Toolkit puts all of that on eight tiles on the watch face.

The app runs on the T-Watch Ultra. For sharp, glare-free pictures, the screenshots in this guide were captured from a desktop simulator running the app's real interface code at the watch's exact 410×502 resolution, with sample tag data. They match the layout on the watch.

At a glance

Tile	Use it to
Scan	Identify any tag: ID, type and what's stored on it. Turn results into a QR code, join a Wi-Fi network, or emulate the tag.
Write	Put a link, text, Wi-Fi login, phone number, SMS, email, contact card, map location or Android app launcher on a tag. Erase or format tags.
Copy	Copy what's stored on one tag onto another.
Dump	Read a tag's entire memory, byte by byte, and save it to the SD card.
Emulate	Make the watch itself act as a tag that phones can tap.
Field	Detect phones, payment terminals and door/transit readers that are sending an NFC field.
RF Lab	See the NFC chip's health: supply voltages, antenna signal, and a live graph that reacts to nearby tags.
History	Review everything you've scanned or dumped, with timestamps.

Getting around

- Tap a tile to open a tool. Tap the **<** button in the coloured header (or press **BOOT**) to go back to the home screen.
- Long results scroll: swipe up and down inside the screen.
- Text boxes open an on-screen keyboard. Tap the tick to close it.
- The watch **vibrates** when a tag is detected, when a write or copy finishes, and when an NFC field appears.
- Leaving a tool always switches the NFC radio off, so nothing keeps transmitting in the background.

Where to hold the tag

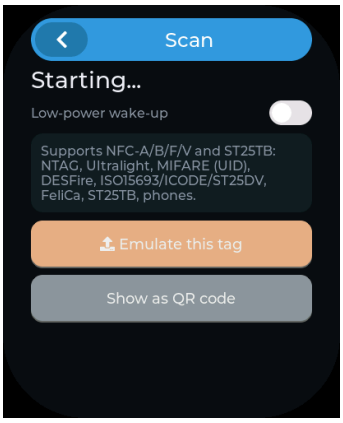
NFC range is only a few centimetres. Hold the tag flat against the watch and move it around slowly until it beeps (vibrates). Once you find the sweet spot, it reads instantly every time. If you're not sure where the antenna is, use **RF Lab**: turn the carrier on, set a baseline, then slide a tag around the watch and watch where the graph jumps. For phones, the NFC coil is usually near the camera on the back.

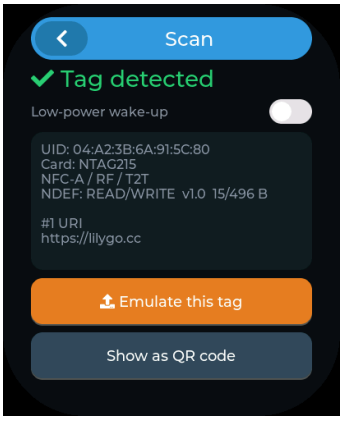
Files and storage

Scan history and saved dumps go in a folder called **/nfc**. If a microSD card is inserted, the app uses it; otherwise it uses the watch's internal storage. History shows which one is in use. Saved dumps are a readable `.txt` plus a raw `.bin`, named by tag ID and time.

Using each tool

Scan





Open **Scan** and hold a tag to the watch. The watch vibrates and shows:

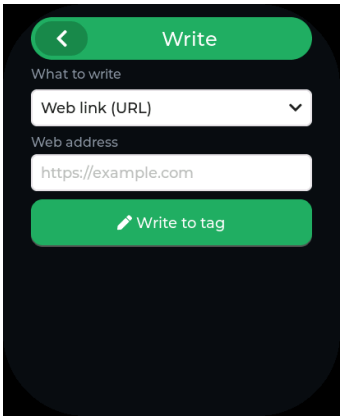
- **UID**: the tag's serial number
- **Card**: the chip model, for example NTAG215 or MIFARE Classic 1K
- **Technology / interface / tag type**: NFC-A, ISO-DEP, Type 2, and so on
- **NDEF**: whether the tag holds standard NFC data, its version, and how full it is
- Each stored record: links, text, Wi-Fi name and password, contact cards, app records

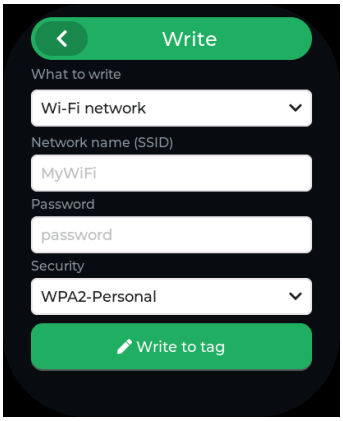
Every scan is saved to History automatically. The buttons under the result use what was found:

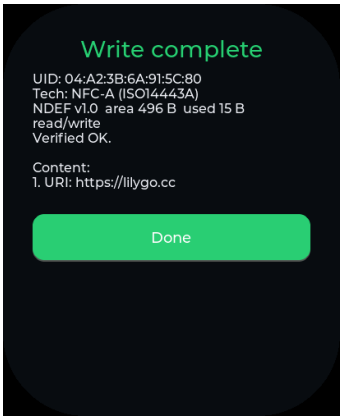
- **Emulate this tag** opens Emulate with the link, text or Wi-Fi details filled in.
- **Show as QR code** shows the link (or UID) full-screen so a phone camera can scan it. Tap to close.
- **Join this Wi-Fi** appears when the tag holds a Wi-Fi login and connects the watch to that network.

Low-power wake-up: when switched on, the chip keeps its field off and only powers up when something disturbs the antenna. That saves battery if you leave Scan open. Reads can take a moment longer to start.

Write





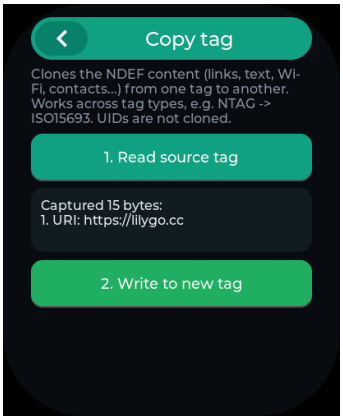


1. Pick **What to write**. 2. Fill in the boxes. 3. Tap **Write to tag** and hold a tag to the watch. The watch writes, reads the data back to check it, then shows **Write complete** or the reason it failed.

Type	What a phone does when it taps the tag
Web link (URL)	Opens the website. If you leave out https://, it's added for you.
Text	Shows the text (or passes it to an NFC app).
Wi-Fi network	Offers to join the network. Choose WPA2, WPA/WPA2 mixed or Open.
Phone call	Opens the dialler with the number.
SMS	Opens a text message with the number and message filled in.
Email	Opens a new email with address, subject and body.
Contact (vCard)	Offers to save a contact with name, phone and email.
Map location	Opens maps at that latitude and longitude.
Launch Android app	Opens (or offers to install) the app with that package name.
Erase tag	Replaces the contents with an empty record. The tag stays reusable.
Format blank tag	Prepares a brand-new, never-used tag so it can hold NFC data.

Brand-new tags are formatted automatically the first time you write to them. If the message is too big for the tag, the watch says so instead of writing part of it.

Copy

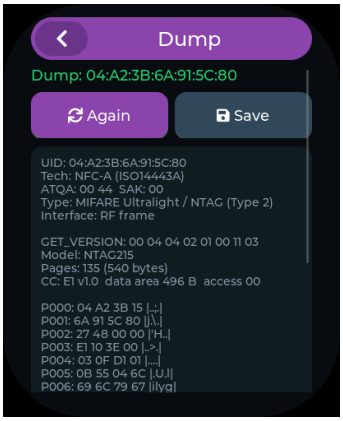


Copies the stored data (links, text, Wi-Fi, contacts) from one tag to another.

- Tap **1. Read source tag** and hold the tag you want to copy. Its contents appear on screen.
- Tap **2. Write to new tag** and hold a different, writable tag. If you hold the original by mistake, the watch tells you and waits for another tag.
- The copy is checked the same way as Write.

The two tags don't have to be the same kind, so you can copy an NTAG sticker onto an ISO15693 tag. Only the contents are copied, not the serial number (UID), which is fixed in each chip at the factory.

Dump

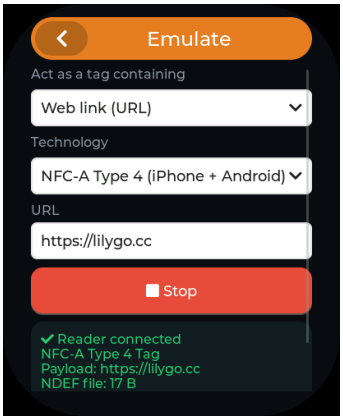




Reads everything the tag will give up and shows it as a hex listing with a text column on the right. It starts listening as soon as you open it. Hold a tag; when it's done, tap **Save** to write it to storage or **Again** for another tag.

Tag	What the dump contains
NTAG21x, Ultralight	Exact model, every page including configuration, decoded NFC data, and NXP's originality signature (proof the chip is genuine NXP).
ISO15693 (ICODE, ST25DV, Tag-it)	System info and every block, including large chips with more than 256 blocks.
ST25TB / SRI	All blocks plus the system block.
FeliCa Lite / Lite-S	The readable (unencrypted) blocks.
Type 4 / DESFire / NTAG 424	The NFC data file and its settings.
Bank and transit cards	Only the names and IDs of the payment apps on the card. No card numbers or transactions.
MIFARE Classic	UID, ATQA and SAK only (see Limits).

Emulate

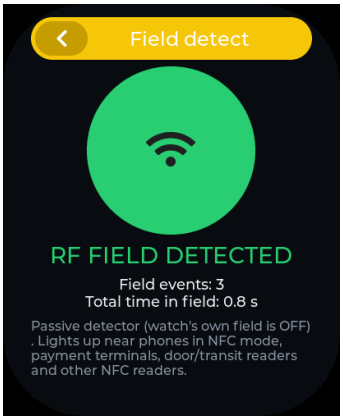


Turns the watch into an NFC tag. Phones that tap it see a link, text or Wi-Fi login, as if it were a sticker.

- Choose what to share, and type it in (or arrive here from Scan's **Emulate this tag** with it filled in).
- Choose the **Technology**: **NFC-A Type 4** works with iPhone and Android; **NFC-F Type 3** (FeliCa) is read well by Android and by phones sold in Japan; **Mixed** offers both.
- Tap **Start emulating** and hold the phone to the watch. The status shows when a reader connects and counts the reads. Tap **Stop** when done.

LilyGO marks the Type 4 emulation in its library as experimental. If a phone won't read it, try Mixed or Type 3.

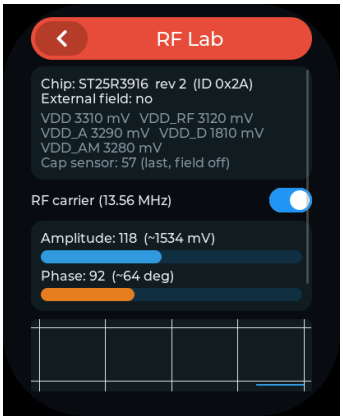
Field detect

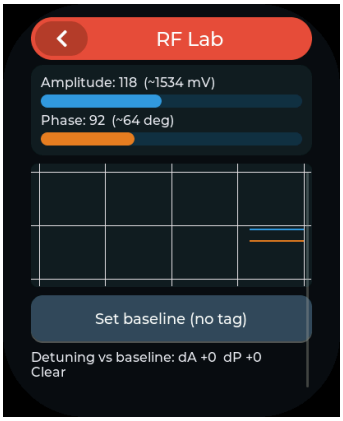


The watch's own radio stays off and listens. When another device sends an NFC field, the circle turns green, the watch vibrates, and the counters go up.

- Find out whether a door, turnstile or payment terminal has an active NFC reader, and where it is.
- Check that a phone's NFC is actually switched on.
- See how often a reader polls (each new field counts as one event).

RF Lab





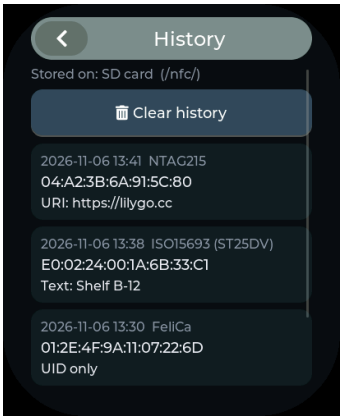
A view into the NFC chip itself, useful when something isn't reading and you want to know why.

Reading	What it means
Chip / revision	Confirms the ST25R3916 is powered and responding. If this is blank, the NFC chip isn't working.
VDD, VDD_RF, VDD_A, VDD_D, VDD_AM	The chip's internal supply rails in millivolts. Around 3.3 V for VDD and VDD_RF is healthy; a low reading usually means a low battery.
External field	YES when another reader's field is present (same as Field detect).
Cap sensor	The chip's built-in capacitance measurement (ST25R3916 only, not the 3916B).
Amplitude / phase	How strong the 13.56 MHz signal at the antenna is, and its phase. Shown live once the carrier is on.

Finding tags by detuning: switch on **RF carrier**, keep the watch away from metal, and tap **Set baseline**. Now bring a tag, a phone or a hand close: amplitude and phase shift and the meter reports how much. This is the same effect the chip uses for its low-power wake-up mode.

If another reader is already transmitting, the chip won't switch its carrier on (collision avoidance); RF Lab tells you when that happens.

History



A list of every tag scanned or dumped, newest first: time, chip type, UID and a one-line summary of the contents. Tap **Clear history** twice to delete it. The file lives at `/nfc/history.tsv`, so you can also open it on a computer.

Times come from the watch's real-time clock. If the clock has never been set, entries show time since power-on instead.

What it can scan, write and emulate

The ST25R3916 speaks every common 13.56 MHz standard: NFC-A (ISO 14443A), NFC-B (ISO 14443B), NFC-F (FeliCa), NFC-V (ISO 15693) and ST25TB. This table shows what the app can do with the tags you're most likely to meet.

Tag / card	Examples	Scan	Read data	Write	Dump	Copy
NTAG210/212/213/215/216	Stickers, key fobs, business cards, NFC implants	Yes	Yes	Yes	Full + signature	Yes
MIFARE Ultralight / EV1	Disposable transit and event tickets	Yes	Yes	Yes	Yes	Yes
MIFARE Classic 1K / 4K / Mini	Older hotel keys, gym and office badges	UID only	No	No	UID only	No
MIFARE DESFire, NTAG 424, ST25TA (Type 4)	Newer access cards, secure tags	Yes	If formatted	If unlocked	NFC file	Yes
ICODE SLIX/SLIX2/DNA, ST25DV/TV, Tag-it (ISO 15693)	Library books, ski passes, industrial tags	Yes	Yes	Yes	Full	Yes
FeliCa Lite / Lite-S	FeliCa NFC stickers	Yes	Yes	Yes	Open blocks	Yes
FeliCa standard	Suica, Octopus and similar transit cards	IDm only	No	No	IDm only	No
ST25TB / SRI512 / SRIX4K	Some transit and laundry cards	Yes	No	No	Full	No
Bank / payment cards (EMV)	Contactless debit and credit cards	Yes	App names	No	App names only	No
Phones	Android / iPhone	Yes	Via Emulate	-	-	-

"Read data" means the standard NFC content (NDEF): links, text, Wi-Fi and so on. A tag can also be locked by its owner; locked tags can be scanned and dumped but not written.

What the watch can pretend to be (Emulate)

Emulated as	Content	Readable by
NFC-A Type 4 tag	Link, text or Wi-Fi	iPhone XS and newer (hold the top edge to the watch) and Android
NFC-F Type 3 tag (FeliCa)	Link, text or Wi-Fi	Android; phones with FeliCa
Mixed A + F	Link, text or Wi-Fi	Whichever the phone polls first

Emulate shares information like a tag would. It can't impersonate a specific card's ID, so it won't open doors or work as a payment or transit card.

Limits, and why

- **MIFARE Classic contents**: the sectors are encrypted with NXP's proprietary Crypto1 cipher. The ST25R3916 has no hardware for it, so only the card's ID, ATQA and SAK are shown.
- **Serial numbers (UIDs) can't be copied**: they're burned into each chip. Copy moves the stored contents only.

- **No permanent lock**: making a tag read-only can never be undone, so the app leaves it out to prevent accidents.
- **Password-protected pages** on NTAG/Ultralight stop the dump at the first protected page; the dump says where it stopped.
- **Payment cards**: only the public list of payment apps is read. Card numbers and transactions are never requested.

Installing the app

Option A: flash the ready-made firmware

Connect the watch by USB-C. On a computer with Python installed:

```
pip install esptool
esptool --chip esp32s3 --port COM5 write-flash 0x0 NFC_Toolkit_TWatchUltra_0x0.bin
```

Use your own port name (Windows: COMx; Mac: /dev/cu.usbmodem...; Linux: /dev/ttyACM0). If the port doesn't appear, hold **BOOT**, tap the reset/power button, release BOOT and try again. Browser flashers such as Espressif's ESP Tool page (Chrome or Edge) work too: add the file at address **0x0** and press Program.

Option B: build from source

- Arduino IDE 2: install **esp32 by Espressif** 3.3.0 or newer from Boards Manager.
- Install **LilyGoLib** and copy everything from **LilyGoLib-ThirdParty** into your Arduino libraries folder.
- Open `NFC_Toolkit.ino`, choose board **LilyGo T-Watch-Ultra**, USB CDC On Boot **Enabled**, keep the default 16 MB partition scheme, and click Upload.

Troubleshooting

Problem	Try this
Nothing happens when I hold a tag	Move the tag slowly around the watch face and back; range is 1–3 cm. Check RF Lab: if the chip line is blank, the NFC chip isn't responding, so restart the watch.
"Tag lost" or a write times out	Keep the tag still until the result appears. Metal surfaces under a sticker block NFC; use anti-metal tags.
Write fails: "read-only"	The tag has been permanently locked. Use another tag.
Write fails: "does not fit"	Shorten the content or use a bigger tag (NTAG216 holds 888 bytes).
iPhone won't read Emulate	Hold the top edge of the iPhone to the watch, screen on and unlocked. If it still won't read, try Mixed.
RF Lab won't turn the carrier on	Another reader (or your phone) is transmitting nearby. Move away from it.
History says "none" for storage	Insert a FAT32 microSD card, or restart so the app can prepare internal storage.

Want detailed logs? Connect to a computer and open a serial monitor at 115200 baud: the NFC library prints each step there.